Xavier Kervyn

Max Planck Institute for Physics, Boltzmannstaße 8, 85748 Garching bei München, Germany
Born in May 2001 in Belgium (Belgian) | xavier.kervyn@mpp.mpg.de | xavierkervyn.github.io/

Current research interests

- ⋆ **String theory**: flat-space and AdS string amplitudes, double copy, motives, tensionless strings, surfaceology
- ⋆ **Top-down holography**: flat-space and twisted holography, AdS/CFT, decoupling limits of string/M-theory
- ⋆ **Mathematics**: algebraic geometry, resurgence theory, intersection theory, celestial and Carrollian symmetries

Academic employment

Max Planck Institute for Physics Garching bei München, Munich, Germany — 10/2025
Doctoral Researcher, MPP QFT + String Theory Groups — –present

- International Max Planck Research School (IMPRS) PhD candidate
- Advised by Stephan Stieberger (string theory) and Johannes Henn (quantum field theory)

Max Planck Institute for Physics Garching bei München, Munich, Germany & remote — 09/2024
Graduate Researcher, MPP QFT + String Theory Groups — 09/2025

- Master's thesis on string amplitudes and flat-space holography
- Advised by Stephan Stieberger (string theory)

Education

Ludwig-Maximilians-Universität München Munich, Germany — 10/2025
PhD in Theoretical Physics — –present

- Advanced coursework in mathematics: algebraic geometry, resurgence theory

Ludwig-Maximilians- & Technische Universität München Munich, Germany — 10/2023
Master of Science in Theoretical and Mathematical Physics (TMP, with Distinction) — 09/2025

- Thesis: *Towards a String Worldsheet Description of Celestial Holography* (1.0, High Distinction, 1 paper)
- Awarded 2 scholarships. Research at **Niels Bohr Institute** & **Max Planck Institute for Physics**

Peterhouse, University of Cambridge Cambridge, United Kingdom — 10/2022
Master of Advanced Study in Applied Mathematics (Part III, with First Class Honours) — 06/2023

- Part III essay: *BMS Symmetries of Gravitational Scattering* (86%, Distinction, published as review)
- Awarded 3 scholarships and 3 prizes. Summer research at **Università degli Studi di Padova**

Ecole Polytechnique Fédérale de Lausanne Lausanne, Switzerland — 09/2019
Bachelor of Science in Physics — 08/2022

- Awarded 2 fellowships and 1 scholarship. Research at **CERN** & **ETH Zürich**

Research stays & long-term academic visits

Niels Bohr Institute Copenhagen, Denmark — 09/2024
Visiting Student, Theoretical High-Energy Physics — 01/2025

- Study project on the superconformal bootstrap of gauge theories with defects
- Supervised by Charlotte Kristjansen and Adam Chalabi

Università degli Studi di Padova Padova, Italy — 07/2023
Visitor, Theoretical Physics

- Research on integrability in superconformal field theories and AdS/CFT
- Supervised by Alessandro Sfondrini and Davide Polvara

CERN CMS collaboration Meyrin, Switzerland 06/2022
Undergraduate Researcher, ETH Zürich High-Energy Physics group 07/2022

– Wrote a Python package for the new high-luminosity CMS electrocalorimeter, detector testing

ETH Zürich Zürich, Switzerland 09/2021
Undergraduate Researcher, Exoplanets and Habitability group 12/2021

– Experimental concept of the LIFE mission, modelled imperfect nulling in the interferometer

Selected awards, grants & scholarships

Bourse d'Études – *Fondation du Domaine de Villette* // CHF 5'000	10/2024
Erasmus+ Scholarship – *Ludwig-Maximilians-Universität München* // €3'000	05/2024
College Prize in Applied Mathematics – *Peterhouse, University of Cambridge* // £300	12/2023
Retrospective Scholarship in Mathematics – *Peterhouse, University of Cambridge* // £125	12/2023
Bourse d'Études – *Fondation du Domaine de Villette* // CHF 5'000	06/2023
Greta Burkill Fund Award – *Peterhouse, University of Cambridge* // £300	03/2023
Bruckmann Fund Award – *Peterhouse, University of Cambridge* // £300	03/2023
Annual Scholarship – *Swiss Study Foundation* // CHF 20'000	11/2022
Excellence Scholarship – *Colbianco Stiftung* // CHF 2'000	08/2022
Fellowship – *Swiss Study Foundation & e-fellows.net* // active fellow until 2026	05/2022
Swiss Mobility Program Grant – *Ecole Polytechnique Fédérale de Lausanne* // CHF 1'500	09/2021
Baccalaureate Merit Award – *Région Provence-Alpes Côte d'Azur* // €400	07/2019

Publications & preprints (Inspire HEP; *:under review) (author ordering is alphabetical in hep-th)

Research articles:

3. X. Kervyn and S. Stieberger, *Resurgence of high-energy string amplitudes*, 2604.07444.
2. X. Kervyn and S. Stieberger, *High-energy string theory and the celestial sphere*, *Journal of High Energy Physics* **09** (2025) 044 [2504.13738]. (based on my master thesis at LMU/TUM)
1. X. Kervyn, D. Polvara and A. Sfondrini, *Thermodynamics of integrable $\mathcal{N} = 2$ theories, squared*, *Journal of High Energy Physics* **09** (2025) 018 [2502.10356].

Reviews:

1. X. Kervyn, *BMS symmetries of gravitational scattering*, *Nuclear Physics B* **1017** (2025) 116948 [2308.12979]. (based on my Part III essay at Cambridge)

Participation to conferences, schools and workshops (*: scheduled, P: poster, T: invited talk, G: gong show)

– Universe+ 2026 Annual Meeting, DAMTP, University of Cambridge, UK [link]		09/2026
– Prospects in Theoretical Physics: Celestial Holography, IAS Princeton, NJ, USA [link]	(G)	07/2026
– Hidden Structures in Multi-loop Amplitudes [link, *online*]		04/2026
– Universe+ School: Exploring Positive Geometry, les Houches, France [link]		03/2026
– Single-valued Periods in Scattering Amplitudes [link, *online*]		01/2026
– Symbology @15, MPI MPP, Munich, Germany [link]		12/2025
– Feynman Calculus and its Applications to Gravity and Particle Physics [link, *online*]		11/2025
– DFG meeting: Modern Foundations of Scattering Amplitudes, BCTP, Bonn, Germany [link]		11/2025
– Geometry and Combinatorics of Scattering Amplitudes, MPI MiS, Leipzig, Germany [link]		10/2025
– From Good Cuts to Celestial Holography, St Anthony's College, Oxford University, UK [link]		07/2025
– Infrared Surprises of Scattering Amplitudes [link, *online*]		05/2025
– School on Asymptotic Symmetries and Flat Holography, G. Galilei Inst., Florence, Italy [link]	(P)	05/2025

- Frontiers in Gravity, Niels Bohr Institute, Copenhagen, Denmark [link] 09/2024
- PhD school: Towards Gravity [link, *online*] 08/2024
- Celestial Holography Summer School [link, *online*] 07/2024
- Strings 2024, CERN, Meyrin, Switzerland [link] 06/2024
- Carollian Physics and Holography, ESI, Vienna, Austria [link] 04/2024
- Celestial 2023, University of Warsaw, Warsaw, Poland [link] 08/2023
- Young Researchers Integrability School & Workshop, Durham University, Durham, UK [link] 07/2023
- Integrability, Dualities and Deformations 2023, Durham University, Durham, UK [link] 07/2023
- Eurostrings 2023, Universidad de Oviedo, Gijon, Spain [link] 04/2023

Invited talks & contributed seminars

Connection problem for exponential periods
UNIVERSE+ online seminar, MPP/MPI MiS/IAS Pinceton/University of Cambridge 10/2026

Ambitwistor strings beyond scattering amplitudes
QFT × String Theory Journal Club, Max Planck Institute for Physics 06/2026

Wall-crossing structure from quantum phenomena to Feynman integrals
Quantum Field Theory Journal Club, Max Planck Institute for Physics 05/2026

Resurgence of high-energy string amplitudes
IMPRS Young Scientists Workshop, Castle Aspenstein 07/2026
String Group Seminar, LMU Munich & Max Planck Institute for Physics 05/2026
Quantum Geometry Group Seminar, Technical University of Munich 05/2026
Theory Seminar, Niels Bohr Institute, Copenhagen 04/2026

Matrix theory reloaded: a BPS road to holography
String Theory Lunch Seminar, LMU Munich & Max Planck Institute for Physics 11/2025

Boundary Carrollian conformal field theories and open null strings
String Theory Lunch Seminar, LMU Munich & Max Planck Institute for Physics 06/2025

Spinor-helicity formalism, twistor theory and scattering amplitudes
Seminar on Scattering Amplitudes, LMU Munich 05/2024

Gauging of discrete higher-form symmetries in non-Abelian Yang-Mills theory
Seminar on Generalized Symmetries in QFT, LMU Munich 12/2023

Gravitational scattering and covariant phase space methods in gravity
DAMTP Part III Seminar Series, University of Cambridge 03/2023

Holography and twistor methods in AdS_5
DAMTP Part III Seminar Series, University of Cambridge 12/2022

Academic service

Peer-review
Physics Reports (1×), Nuclear Physics B (1×) 2026-...

Technical skills

Languages: French (native speaker); English, German (full work proficiency); Danish, Norwegian (basics)
IT: Object-oriented programming (C++, Python), `Mathematica`, MATLAB, LaTeX (GitHub+VSCode/Overleaf)

Academic references (thesis advisory committee)

- **Dr. Stephan Stieberger**
  Permanent Staff Member & Research Group Leader, String Theory, Max Planck Institute for Physics
- **Prof. Dr. Johannes Henn**
  Director & Department Head, Quantum Field Theory, Max Planck Institute for Physics
- **Prof. Dr. Niels Obers**
  Professor & Institute Research Deputy Head, Niels Bohr Institute